UNIT 6 · TOPIC 6

Mechanics of Traction Devices and Traction Theories

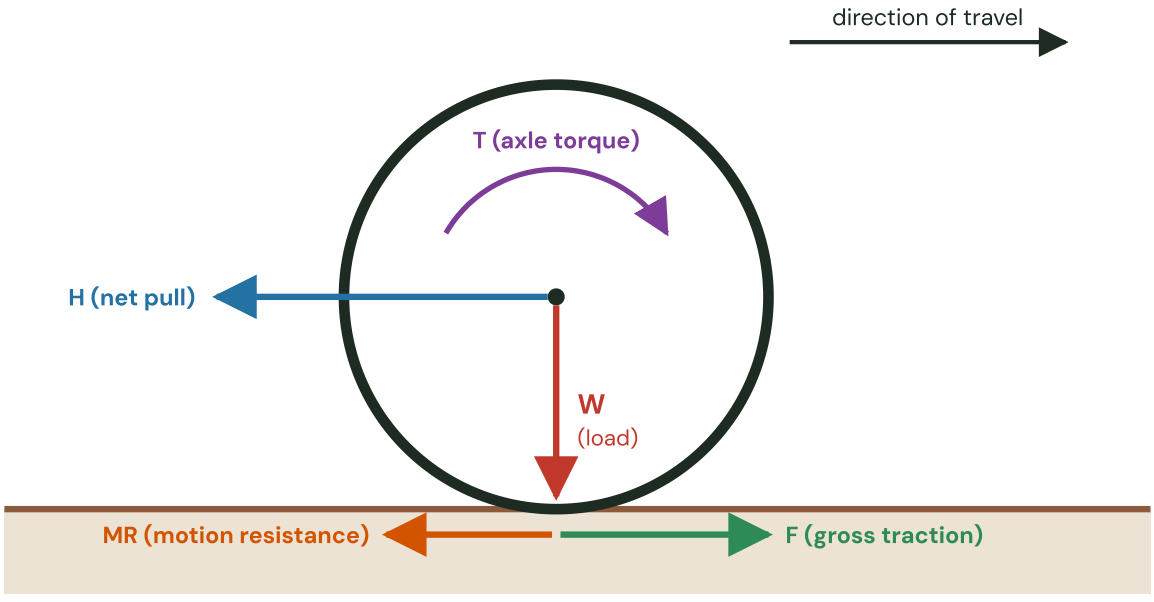
Syllabus topic: Mechanics of traction devices and traction theories

Core definitions (Liljedahl p. 243)

Term	Definition
TRACTIVE EFFICIENCY (TE)	Output power ÷ input power of the traction device — how well axle torque becomes linear drawbar pull; lowered by rolling resistance, slip, tyre friction/deflection and steering
NET TRACTION COEFFICIENT μ	Net pull ÷ dynamic normal load (H/W)
GROSS TRACTION COEFFICIENT μ_g	F/W
MOTION-RESISTANCE RATIO ρ	Rolling-resistance force ÷ normal load (MR/W)

$$\mu = \mu_g - \rho \text{ (net = gross − motion resistance)}$$

Forces on a driving wheel (pp. 243–246)



H = F − MR • S = 1 − Va/Vt • μ = H/W, μ_g = F/W, ρ = MR/W, μ = μ_g − ρ

Forces on a driving wheel, after Wismer & Luth 1974 / Liljedahl Ch. 10 (redrawn — the source figure's torque arrow turned against travel)

- A driving wheel carries dynamic load W and receives axle torque T. At the contact patch soil shear develops a GROSS traction force F; motion resistance MR opposes travel.

- The useful output is the NET PULL (drawbar): **H = F − MR**.

Force state (Wismer & Luth)	Boundary
BRAKED	—
TOWED wheel	Zero axle torque — separates braked from driven
DRIVEN	—
SELF-PROPELLED	Zero pull; torque just overcomes MR — separates driven from driving (p. 245)
DRIVING	—

Weight transfer and traction

- Drawbar pull transfers load off the front axle onto the rear ($\Delta R = P \cdot h / L$ — Unit 4, Topic 9a), raising the rear-wheel reaction and hence the maximum pull of a 2WD tractor.
- Liljedahl's proviso (p. 243): raising the rear-wheel reaction increases traction ONLY IF the soil is strong enough and sinkage does not limit it — the whole story of wet-soil traction (Topic 2: $\varphi \to 0$).

Soil-thrust (Mohr–Coulomb) basis

$$H_{max} = A \cdot c + W \cdot \tan \varphi \text{ (cohesion term — area-dependent; friction term — load-dependent)}$$

Soil	Thrust	What helps
Sand ($c \approx 0$)	$\propto$ load	Ballast
Saturated clay ($\varphi \approx 0$)	$\approx A \cdot c$, independent of load	More contact area — tracks, large tyres; ballast only sinks the wheel

Two routes to traction prediction (pp. 246–252)

1. **ANALYTICAL** Mohr–Coulomb / Bekker theory — needs c, φ and the k values; Liljedahl notes that convenient evaluation of c and φ in the field is difficult (p. 246).
2. **DIMENSIONAL ANALYSIS using CONE INDEX** — the practical, dominant method (Topic 8).

Practice MCQs

Q71. Traction is defined as:

(a) The braking force

(b) The driving force developed by a wheel/track/traction device

(c) Rolling resistance

(d) Tyre pressure

Q72. Tractive efficiency (TE) is the ratio of:

(a) Pull to weight

(b) Output power to input power of the traction device

(c) Slip to sinkage

(d) Torque to speed

Q73. Net traction coefficient μ equals:

(a) Gross force/torque

(b) Net pull ÷ dynamic normal load

(c) Slip%

(d) Rolling radius

Q74. Motion-resistance ratio ρ is:

(a) MR ÷ normal load
(b) Pull ÷ MR
(c) Slip
(d) Sinkage/width

Q75. Net and gross traction coefficients are related by:

(a) μ = μ_g + ρ
(b) μ = μ_g − ρ
(c) μ = μ_g × ρ
(d) μ = ρ − μ_g

Q76. Net pull H of a driving wheel equals:

(a) F + MR
(b) F − MR (gross traction minus motion resistance)
(c) W − F
(d) T/r

Q82. A towed (unpowered) wheel has:

(a) Maximum torque
(b) Zero axle torque
(c) Zero load
(d) Maximum pull

Q83. For a self-propelled wheel condition:

(a) Pull is zero, torque overcomes motion resistance
(b) Torque is zero
(c) Slip is 100%
(d) Load is zero

Q84. Weight transfer under drawbar pull on a 2WD tractor:

(a) Reduces rear traction
(b) Increases rear-wheel reaction and maximum pull
(c) Has no effect
(d) Lifts the rear

Q85. Increasing rear-wheel reaction increases traction only if:

(a) The soil is strong enough and sinkage does not limit it
(b) Always
(c) Speed is high
(d) Tyre is bald

Q86. Maximum soil thrust of a device is H =

(a) Ac + W tanφ
(b) A/c
(c) W/A
(d) c − W

Q87. In dry sand (c≈0), traction is improved by:

(a) Adding weight (thrust ∝ load)
(b) Reducing weight
(c) Wetting
(d) Nothing

Q88. In saturated clay (φ≈0), adding ballast mainly causes:

(a) More traction
(b) Sinkage/compaction with little gain
(c) Less slip
(d) Higher TE

Answers

Q71 — (b) The driving force developed by a wheel/track/traction device

Q72 — (b) Output power to input power of the traction device — *Liljedahl p. 243 — TE = output/input power.*

Q73 — (b) Net pull ÷ dynamic normal load

Q74 — (a) MR ÷ normal load

Q75 — (b) μ = μ_g − ρ — *μ = μ_g − ρ.*

Q76 — (b) F − MR (gross traction minus motion resistance) — *H = F − MR.*

Q82 — (b) Zero axle torque

Q83 — (a) Pull is zero, torque overcomes motion resistance

Q84 — (b) Increases rear-wheel reaction and maximum pull

Q85 — (a) The soil is strong enough and sinkage does not limit it

Q86 — (a) Ac + W tanφ — *Mohr–Coulomb thrust (Topic 2).*

Q87 — (a) Adding weight (thrust ∝ load) — *Sand c ≈ 0 — ballast helps.*

Q88 — (b) Sinkage/compaction with little gain — *Wet clay φ ≈ 0 — ballast sinks.*

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Wismer & Luth (1974), Off-road traction prediction for wheeled vehicles, Trans. ASAE; Freitag (1965) mobility-number dimensional analysis; Bekker pressure–sinkage theory; ASAE S296.2 (slip/rolling radius), S313.2 (cone index); NPTEL traction lectures.